



SingleViper: A Novel Protein-Based Cell-Type Annotation Workflow

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Guiding Question

How can we improve cell type annotation in single-cell RNA-seq data by combining VIPER-inferred protein activity with SingleR reference-based classification?

Plain Text Summary

Purpose: SingleViper is a computational pipeline for single-cell analysis that integrates gene regulatory network inference with protein-level activity estimation for cell-type classification.

Core Components:

- **ARACNe3** – infers gene regulatory networks and regulons (transcription-factor target sets).
- **VIPER** – estimates transcription-factor protein activity from regulons.
- **Custom SingleR layer** – classifies cells based on protein-activity profiles rather than only gene expression.

Key Advantage: Uses regulon-based protein activity instead of raw or log-normalized gene expression, reducing noise from data sparsity and technical dropouts.

Outcome: Achieves improved cell-type classification accuracy through this higher-level representation.

Implementation: Written in R as modular scripts designed for scalability and deployment on high-performance computing (HPC) clusters.

Methods

Data Input & Preprocessing

- Raw UMI count matrices (query & reference)
- Quality control, log-normalization, and Z-score transformation
- Down-sampling by UMI depth

Gene Regulatory Network Inference

- **ARACNe3** to infer transcription factor regulons
- Supports custom network pruning and parallel computing

Protein Activity Inference

- **VIPER** converts regulons into per-cell protein activity scores

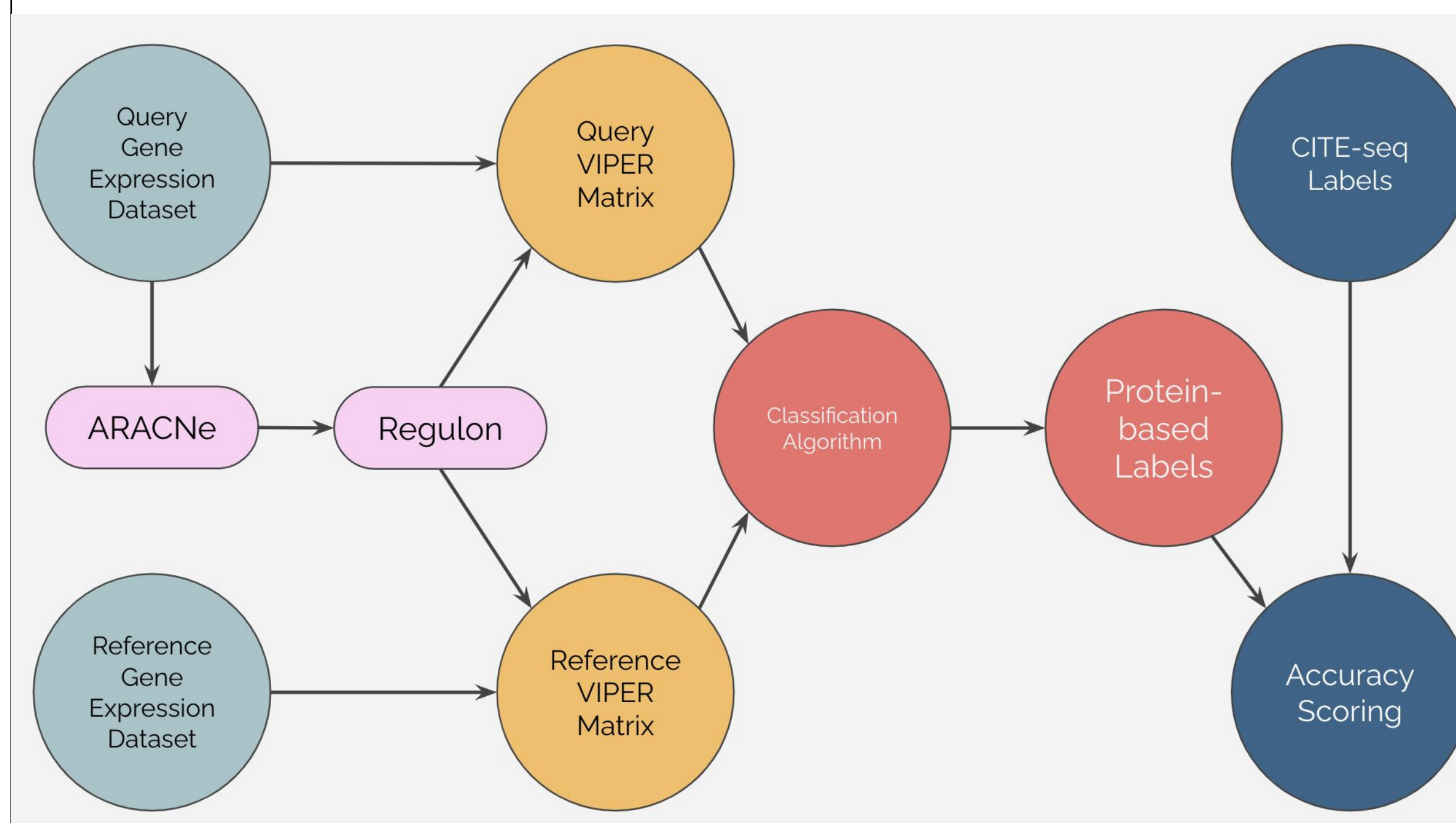
Reduces noise from gene-level dropouts

Cell Type Annotation

- Custom **SingleR** implementation compares VIPER protein activity to reference datasets (Blueprint, HPCA, CITeseq)
- Includes correlation-based and machine-learning similarity metrics

Integration & Visualization

- Combines VIPER activity with Seurat for dimensionality reduction (UMAP)
- Outputs cell type labels, confidence scores, and QC plots (e.g., UMI histograms, activity heatmaps)



Contact me at nlm2166@columbia.edu!

Anticipated Results

Improved Accuracy: VIPER-based SingleR classification should achieve higher concordance with curated cell types versus expression-only approaches, especially in rare or low-depth populations.

Robust to Sparsity: Will maintain high annotation performance on down-sampled datasets, outperforming log-normalized expression inputs.

Flexible References: Compatible with multiple reference atlases (Blueprint, HPCA, CITeseq) and custom datasets.

